

$$\lim_{x \rightarrow 0} (x \cdot \sin(\frac{1}{x})) \rightarrow \left. \begin{array}{l} \text{Subst.} \\ \text{Direta} \\ \text{n\~ao funciona} \end{array} \right\} \nexists \lim_{x \rightarrow 0} \sin(\frac{1}{x})$$

$f(x) = x \cdot \sin(\frac{1}{x})$

**Teorema do**  
**CONFRONTO**  
**(1º)**

$$\rightarrow \overset{1^\circ}{-1} \leq \sin(\frac{1}{x}) \leq \overset{1^\circ}{1}$$

$$-x \leq \underbrace{x \cdot \sin(\frac{1}{x})}_{f(x)} \leq x$$

On regra,  $-x \leq f(x) \leq x$

$x \rightarrow 0 \Rightarrow -x \rightarrow 0 \wedge x \rightarrow 0$

$|f(x)| \leq |x|$

ENTÃO,  $f(x) \rightarrow 0$ . On regra,

$$\lim_{x \rightarrow 0} f(x) = 0$$

$$\lim_{x \rightarrow 3^+} \frac{1}{x-3} = \frac{1}{3^+ - 3} = \frac{1}{0^+}$$

DIVISÃO DE 1 POR N° MUITO Pequ. e positivo  $= +\infty$

SEMPRE  
Aprox. por  $x > 3$

$x \rightarrow 3^+ \sim$

3 (3,1, 3,01, 3,001, 3,0001, ...)

$$\lim_{x \rightarrow 3^-} \frac{1}{x-3} = \frac{1}{0^-} \left\{ \begin{array}{l} \text{2 DIVIDIDO} \\ \text{POR N}^\circ \\ \text{"MUITO PEQ."} \\ \text{negativo} \end{array} \right. = D = -\infty$$

$x \rightarrow 3^-$



$(2,9, 2,99, 2,999, \dots)$

$$\lim_{x \rightarrow 3^-} \left( \frac{-2}{x-3} \right) = \frac{-2}{0^-} = +\infty$$

$$\lim_{x \rightarrow 3^-} \left( \frac{x}{x-3} \right) = \frac{3^-}{0^-} = -\infty$$

$$\lim_{x \rightarrow 3^-} \left( \frac{-\frac{1}{x-3}}{\frac{x-3}{x-3}} \right) = \frac{\frac{-1}{0^-}}{0^-} = \frac{\frac{1}{0^+}}{0^-} = \frac{+\infty}{0^-} = -\infty$$

$f(x)$

Subst. Direta (ok)

$$\frac{-\frac{1}{x-3}}{\frac{x-3}{x-3}} = -\frac{1}{x-3} \cdot \frac{1}{x-3} = -\frac{1}{(x-3)^2}$$

$$\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} \left( -\frac{1}{(x-3)^2} \right) = \frac{-1}{(0^-)^2} = \frac{-1}{0^+} = -\infty$$